MIL-I-37588
29 July 1977

MILITARY SPECIFICATION

# INTRAUTERINE DEVICE, CONTRACEPTIVE, PLASTIC, RADIOPAQUE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of sterile, radiopaque, intrauterine device and sterile insertion unit suitable for contraceptive purposes.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-S-36586 - Sterilization Strip Test, Bacterial Spore.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 19042.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material shall be as indicated herein and of the highest quality suitable in all respects for the purpose intended.

3.1.1 Indwelling device. The indwelling device shall be fabricated of a suitable polypropylene homopolymer with barium sulfate added to render it radiopaque. The tail thread shall be fabricated from a monofilament plastic. The delivery tube assembly shall be fabricated of a suitable flexible plastic material with appropriate centimeter marking to measure depth of penetration. All plastic components of the indwelling device shall meet the requirements of the United States Pharmacopeia (USP) for class VI plastic when tested as specified in 4.4.1. Copper on the vertical limb shall be pure virgin electrolytic copper wire.



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3.1.2 Barium sulfate. The intrauterine device shall contain USP grade barium sulfate to render the device visible to X-ray examination. Shall be in accordance with 4.5.3.

3.1.3 Radiopacity. The intrauterine device shall be tested in accordance with 4.5.3.1.

3.2 Style, design, and dimensions. The intrauterine device shall be sterile, radiopaque and suitable for the control of conception. The shape shall approximate the Arabic numeral 7 and shall retain its memory and return to the Arabic numeral 7 configuration after insertion. A pure, virgin electrolytic copper wire shall be coiled around the vertical limb to provide the surface area approximately 200 square millimeters. The proximal end of the device shall have a suitable plastic thread to facilitate inspection and removal. The intrauterine device shall be in strict accordance with figure 1.

3.3 Sterility. The intrauterine device and insertion unit shall be sterile when tested with a biological indicator as specified in 4.5.2.

3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

(a) Item Identification.
(b) National Stock Number.
(c) Contract or purchase order number.
(d) Sterility lot or control number.
(e) Date of sterilization.
(f) Quantity of production in sterilizer lot.
(g) Type of sterilization (gas, radiation, heat, steam, etc.).
(h) Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
(i) Test organism(s) and population density used.
(j) Test medium incubation temperature, incubation time, and dates of tests.
(k) Number of test units.
(l) Statement that samples comply with sterility tests.
(m) Signature of head of laboratory or a designated qualified alternate.



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When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility (lot) control number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.4 Animal implantation. When tested as specified in 4.5.1.1, the intrauterine devices shall show no evidence of toxicity or adverse tissue reaction upon gross and microscopic examination.

3.5 Certificates of biological tests and animal implantation tests. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested in accordance with methods described in the USP and found to have passed the biological tests and the animal implantation test. This certificate shall include the specific tests performed and the test results. The record of the animal implantation test and its result shall be certified by the pathologist who performed or supervised performance of the animal implantation test. For destination inspection, certificates shall be submitted by the contractor directly to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via government inspector.

3.6 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all sterility, biological, and animal implantation tests performed on the intrauterine devices. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.



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3.7 Instructions. Each intrauterine device shall be provided with a detailed instruction sheet containing the following minimum information: Description of the device; indications and contraindications for use; time of insertion; techniques for correct insertion and removal of the device with appropriate anatomy illustrations; possible side effects; patient instructions; warnings as to copper allergies including Wilson's disease; specification recommendations shall be made as to the duration of use as to when the copper has been expended in reference to contraceptive action and when the device should be replaced. All instructions shall be in accordance with the regulations and requirements of the Food and Drug Administration.

3.8 Labeling. Physician and patient product labeling shall provide information on the nature of the IUD, its side effects, contraindications and potential hazards, and the need for continued medical supervision in accordance with those of Food and Drug Administration.

3.9 Workmanship. Workmanship shall be first class throughout. The intrauterine devices shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.



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4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the plastic or the barium sulfate may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one intrauterine device and inserter.

TABLE I. Sampling for examination.

| Defect category | Inspection level | AQL (percent defective) |
|---|---|---|
| Major | S-2 | 1.0 |
| Minor | S-1 | 1.5 |

4.2.2 For tests.

4.2.2.1 For sterility test. Sampling for the sterility test shall be conducted as indicated in the USP for non-parenteral devices.

4.2.2.2 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. Where feasible, the same sample shall be used for the determination of two or more test characteristics. The unit of product shall be one intrauterine device and inserter.

TABLE II. Sampling criteria for tests.

| Component | Characteristic | Require-ment | Test procedure | Inspection level | AQL (percent defective) |
|---|---|---|---|---|---|
| Unit of product | Material-class VI | 3.1.1 | 4.5.1 | 2/ | 1/ |
| IUD | Barium | 3.1.2 | 4.5.3 | S-2 | 1.0 |
| IUD | Radiopaque test | 3.1.3 | 4.5.3.1 | S-2 | 1.0 |
| Unit of product | Dimensions | 3.2 | ------- | S-2 | 1.0 |



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TABLE II. Sampling criteria for tests - continued

| Component | Characteristic | Requirement | Test procedure | Inspection level | AQL (percent defective) |
|---|---|---|---|---|---|
| Unit of product | Sterility | 3.3 | 4.5.2 | S-2 | 1/ |
| Unit of product | Animal implantation | 3.4 | 4.5.1.1 | S-1 | 1/ |

1/ Acceptance number shall be zero.
2/ Each lot of product need not be biologically tested provided evidence is furnished that samples of the formulation and identical method of processing had previously been tested, that they comply with USP requirements for class II plastics and that they are the same as being offered in this contract.

4.2.3 Raw materials. Sufficient quantity of each lot of barium sulfate and each lot of each plastic used in the construction of the intrauterine device shall be selected for chemical analysis.

4.3 Examination. The intrauterine device and inserter shall be examined to determine compliance with all the requirements in accordance with MIL-STD-105 and the classification of defects specified in tables I and III. Examination, however, shall not be restricted to the classified possible defects listed in table III. Any intrauterine device and inserter in the sample containing one or more defects shall be rejected. If the number of defective devices in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects.

| Categories | Defects |
|---|---|
| Critical | None defined. |
| Major | |
| 101 | One or more components missing. |
| 102 | Intrauterine device not free of cracks, burrs, or slivers. |
| 103 | Surfaces of device, plunger or tube not smoothly finished, or not free of flash, or feathering. |



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TABLE III. Classification of defects - continued

| Categories | Defects |
|---|---|
| Major | |
| 104 | Tube lumen not free of obstructing material, burrs, chips, dirt, or other foreign material. |
| 105 | Plunger does not make a free, sliding fit within tube. |
| 106 | Instruction sheet, missing, incomplete, or illegible. |
| Minor | |
| 201 | Poor workmanship (not resulting in a major defect). |

4.4 Packaging inspection. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE IV. Packaging inspection.

| Examination | Defect |
|---|---|
| Containers | Not as specified. |
| Quantity | Quantity in interior packages or shipping containers not as specified. |
| Packaging & packing materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted. |
| Marking | Omitted, incorrect, illegible, improper size, location, sequence, or method of application. |

4.4.1 Examination of closure, waterproofing, and banding of containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing and banding shall be in accordance with the appendix to the box specification.



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4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Biological tests for class VI plastic. The intrauterine device shall be tested in accordance with the requirements of the USP for class VI plastic.

4.5.1.1 Animal implantation test. Samples of the finished intrauterine device shall be tested in accordance with the USP for Implantation Test, Plastic Containers, for class VI plastic, except after "72 hours" add "and not to exceed 144 hours" and delete "Examine macroscopically the area of the tissue surrounding the center portion of each implant strip. Use a magnifying lens if necessary" and substitute: "The muscle tissue surrounding the implants shall be examined, both grossly and microscopically, by a pathologist certified by the American College of Veterinary Pathologists."

4.5.2 Sterility test. The intrauterine devices, and insertion units shall be tested for sterility in accordance with the method specified in the USP for nonparenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.5.2.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table II, Sampling criteria for tests.

4.5.2.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

| Type of Sterilization | Organism | Incubation temperature |
|---|---|---|
| Steam | B. stearothermophilus | 55° - 65°C |
| Dry heat | B. stearothermophilus, or | 55° - 65°C |
| | B. subtilis var. niger (or globigii) | 30° - 35°C |
| Gas | B. subtilis var. niger (or globigii) | 30° - 35°C |
| Radiation | B. pumilus | 30° - 35°C |

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.



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4.5.2.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.5.2 Sterility test and 4.5.2.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.5.2.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.5.2.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Request for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.5.2.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.5.3 Barium sulfate. Accurately weigh a 6.5 gram sample of the IUD plastic and proceed in accordance with the USP method of assay for barium sulfate.

4.5.3.1 X-Ray opacity test. The intrauterine device shall be placed on an unexposed X-ray plate and covered with an aluminum ladder ranging in thickness from at least 1/8" to 3/4" in steps of 1/8". Using an appropriate exposure such that the steps of at least 1/4", 3/8", 1/2" and 5/8" are clearly delineated from each other, one should note the insert appearance.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing thirty intrauterine devices, as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Individual container. Each intrauterine device shall be mounted on a paper board stiffener or other suitable material in such a manner as to prevent distortion to the assembly. The unit along with instructions for use shall be further protected by being packaged in a sealed peel type container of appropriate



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size and design. The container shall be capable of maintaining sterility of the contents until manually opened and shall not be resealable.

5.1.1.1.2 Unit package. Each unit of issue (thirty intrauterine devices) shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of PPP-B-636.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Five (5) units of issue shall be packed in an exterior container designed for type 2 load and constructed in accordance with PPP-B-636, class weather resistant, graded V3c. Closure, waterproofing and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Five (5) units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual container. In addition to any commercial marking, each individual container shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act. Marking shall include the lot (control) number and the legend: "STERILE, UNLESS PACKAGE IS OPENED OR DAMAGED" or similar statement.

5.3.2 Unit package. Each unit package shall be marked as specified in MIL-STD-129. Marking shall include the lot (control) number, contract or purchase order number, name of contractor, and the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat sterilization) shall be shown following the level of packaging and date of packaging. Example: A-8/76-G. Final sterilization shall be on or after indicated packaging date.

6. NOTES



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6.1 Intended use. The intrauterine devices are intended for optimum uterine placement, fit, tolerance and retention for effective contraception.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 National Stock Number (NSN). This specification is intended to cover the following item appearing in the Federal Supply Catalog:

| National Stock Number | Item Identification |
|---|---|
| 6515-01-003-5345 | *INTRAUTERINE DEVICE*, Contraceptive, Plastic, Radiopaque, 30's |

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
Defense Logistics Agency - DM

Project No. 6515-2037

☆U.S. GOVERNMENT PRINTING OFFICE: 1977—703-020/5398



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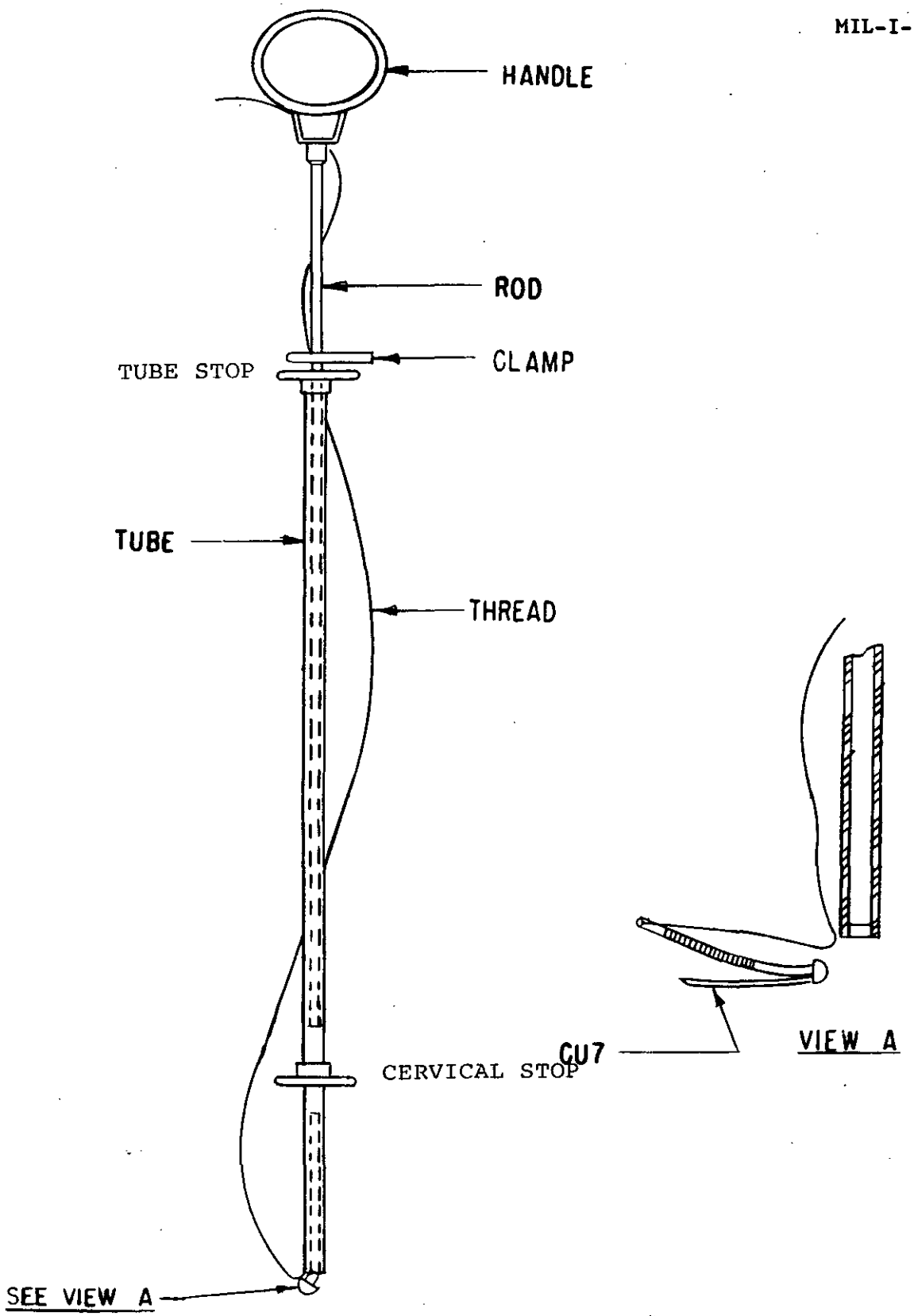



FIGURE 1. Intrauterine device, contraceptive, plastic, radiopaque.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

*INSTRUCTIONS:* The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-I-37588 INTRAUTERINE DEVICE, CONTRACEPTIVE, PLASTIC, RADIOPAQUE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT *CONSIDERED TOO RIGID*

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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